Employee Attrition & Workforce Analytics using Python

Exploratory Data Analysis Report

Dataset size: 1,470 rows × 35 columns.

Overall attrition rate: 16.12% (237 employees marked Yes).

1. Objective and business questions

This analysis explores employee attrition by department, job role, overtime, income, satisfaction, distance from home, age, work-life balance, and tenure. The objective is to identify descriptive patterns that may help guide further HR investigation.

2. Data quality

Exact duplicate rows detected: 0. Missing values detected: 0. Constant columns: EmployeeCount, Over18, StandardHours. Potential outliers are flagged using the 1.5×IQR rule and are not automatically removed.

3. Key findings

• Overall attrition: 16.12% (237 leavers out of 1470 employee records).

• Highest observed attrition rate by Department: Sales (20.6%; n=446). Check group size before drawing conclusions.

• Highest observed attrition rate by JobRole: Sales Representative (39.8%; n=83). Check group size before drawing conclusions.

• Highest observed attrition rate by OverTime: Yes (30.5%; n=416). Check group size before drawing conclusions.

• Highest observed attrition rate by BusinessTravel: Travel_Frequently (24.9%; n=277). Check group size before drawing conclusions.

• Median MonthlyIncome: leavers=3202.00; stayers=5204.00.

• Median Age: leavers=32.00; stayers=36.00.

• Median DistanceFromHome: leavers=9.00; stayers=7.00.

• Median JobSatisfaction: leavers=3.00; stayers=3.00.

• Median WorkLifeBalance: leavers=3.00; stayers=3.00.

• Median YearsAtCompany: leavers=3.00; stayers=6.00.

4. Recommendations

• Review overtime workload and staffing practices; compare attrition rates and group sizes before changing policy.

• Investigate high-attrition job roles with HR interviews and role-specific retention analysis.

• Evaluate compensation bands and career progression using pay equity and job-level context rather than salary alone.

• Review job satisfaction and work-life balance feedback alongside manager and team context.

• Use voluntary, privacy-respecting employee feedback and repeat analyses over time if dated records become available.

• Do not use these descriptive patterns to make individual employment decisions; validate findings and assess fairness.

5. Limitations

This is a cross-sectional, observational dataset. Findings describe associations and do not prove causality. The dataset may not represent every employer or workforce. Some fields are ordinal ratings, so numerical summaries should be interpreted cautiously. There is no suitable date field for time-series analysis. Attrition-rate comparisons should be reviewed with group sizes and business context.

6. Reproducibility

All results are generated from the CSV loaded in this notebook. See the notebook for cleaning checks, plots, group-by tables, correlation analysis, and outlier diagnostics.